

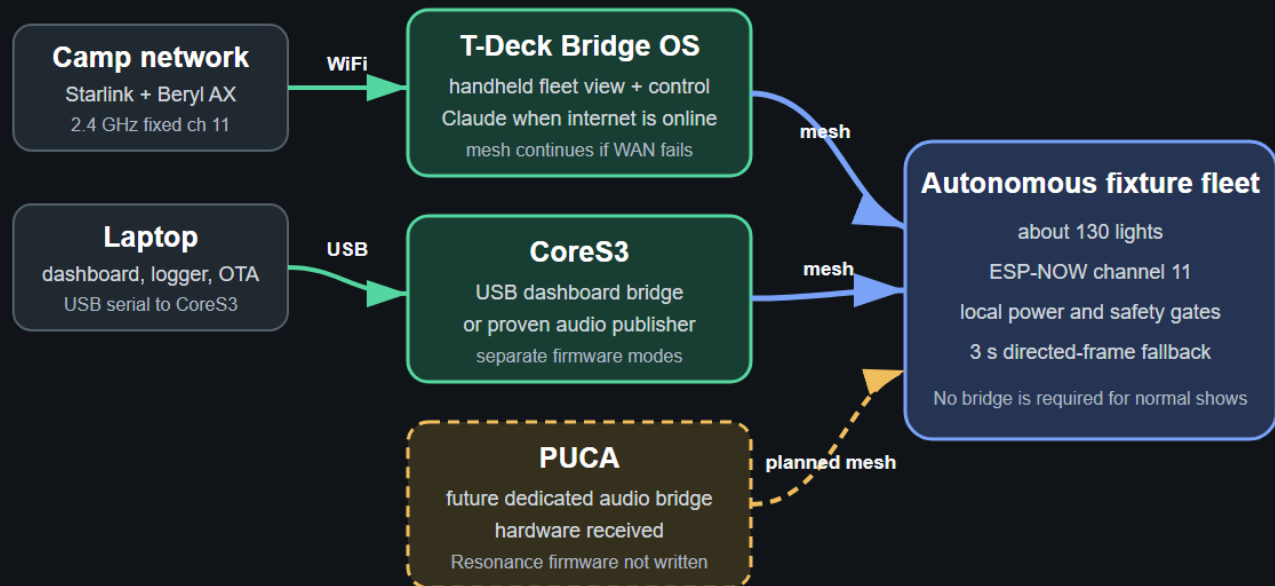
Resonance Bridge Field Manual

Bridge OS, CoreS3, and PUCA without the archaeology

Living document: 2026-08-24

For field operators, lighting crew, IT support, and developers on call

Three bridge tools, one channel-11 fixture fleet



Operational rule: run one artistic frame publisher at a time; state-changing maintenance has one declared operator.

The lights are autonomous. Bridges observe or temporarily direct them; local battery, lifecycle, and actuator safety always win.

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Read this first

These six rules prevent most field mistakes:

1. The fleet is on ESP-NOW channel 11. A T-Deck may use WiFi and the mesh at the same time only when the 2.4 GHz access point is also on channel 11.
2. A callsign is friendly; the six-hex short MAC is identity. Say and record both for any state-changing work: Ponyta [F2B7DC], for example.
3. Check the live count before a fleet action. A sleeping or silent fixture cannot hear a best-effort broadcast. seen is history; live is reachable now.
4. Dark and sleep are different. Dark keeps the radio awake and draws much more power. Sleep turns off the rails and radio and cannot be cancelled over the air until the timer wakes the fixture.
5. Leaving LEDs or Patterns does not stop the look. Use the app's Stop button. A lost direct stream falls back to autonomous output in about three seconds.
6. Use one live artistic publisher at a time. T-Deck LEDs, T-Deck Patterns,

CoreS3 audio, and future PUCA audio all use directed frames. Two publishers can visibly fight because artistic control is intentionally last-writer-wins.

Security note: fleet commands are not yet authenticated on the radio. Keep the T-Deck in trusted hands. Physical possession is an operating mitigation, not a cryptographic security property.

For OTA, USB flashing, reboot, profile/channel persistence, or any other NVS mutation, use one declared operator across all bridges and laptops. Follow [Firmware artifact identity and shared-bench handoff](#).

Which bridge should I use?

Need	Use	Why
Walk the site, check health, identify a light, or set a temporary look	T-Deck Bridge OS	Handheld, mesh-native, works without a laptop; Claude is optional
See the complete host dashboard or record serial telemetry	CoreS3 normal bridge + laptop	Proven USB bridge for the dashboard and logger
Run audio-reactive lights today	CoreS3 + Module Audio + RODE	The only audio path already hardware-validated on fixtures
Use Cambium's binary serial transport	CoreS3 Cambium build	Separate firmware mode; not compatible with the text dashboard
Run the future dedicated performance audio box	PUCA, after bring-up	Better final audio/control hardware, but Resonance firmware is not written yet
Run the ordinary autonomous show	No bridge required	The fixture fleet is designed to keep working without infrastructure
Update fixture firmware	Laptop + shared-WiFi OTA tools	A bridge can request exact-target maintenance, but it is not the OTA uploader

Status at a glance

The words below are deliberately conservative:

- Working means hardware evidence exists for the main path.
- Canary means the feature is in the flashed image, but its explicit field

acceptance matrix is still open.

- Planned means do not expect a working feature.

Device or feature	Status on 2026-08-24	Practical meaning
T-Deck mesh, launcher, touch, trackball, keyboard, GPS, channel guard	Working	Hardware-probed on T-Deck 8EB508
T-Deck Claude, Fleet, LED Studio, Sleep / Dark, single Knock, CA Studio	Working with limits	Broad hardware or field smoke tests passed; read the app notes below
T-Deck Health and Schedule	Canary	Broad physical smoke passed; full color/off-air/override matrices remain open
T-Deck callsigns, Patterns v1, RF Diagnostics, full-fleet Knock roll	Canary	Current image is flashed; deliberate named-canary checks remain open
T-Deck Locate, detailed sensor samples, microphone Patterns, voice	Planned	Locate is a visible placeholder; the others have no finished operator path
CoreS3 normal dashboard bridge	Working	Channel-11 mesh and dashboard integration proven
CoreS3 Module Audio bridge	Working	RODE path and three-fixture stale fallback proven
PUCA Resonance bridge	Planned	Hardware received; factory firmware is not Resonance firmware

The current T-Deck unit is short ID 8EB508, full MAC 44:1B:F6:8E:B5:08. The callsign-aware binary recorded on 2026-08-24 is 1,550,224 bytes with SHA-256 3026593615bd58304c2a6b8893bf4f92cd8f9f92211f9222a5a28517fedf6e32. A COM port is an observation, not identity; COM152 was the observed port at that flash, not a permanent name.

Part I: T-Deck Bridge OS

The one-minute startup

1. Power the T-Deck and wait for the launcher.
2. Read the top line. Confirm ch11, confirm the mesh live/seen count is plausible, and note whether the left side says online, mesh-only, connecting, or guard.
3. Open Health. A mostly grey screen immediately after boot can be normal; give sleeping fixtures time to report.
4. If you will command a fixture, name the intended callsign and short ID before touching the control.

5. If you need Claude, confirm WiFi is online. Every mesh app still works if internet is down.

If the launcher says GUARD, do not change the fleet channel. Fix the 2.4 GHz access point so it is on channel 11, then use `serial wifi retry`.

Screen and controls

The T-Deck has three useful input methods:

- Touch: fastest with bare hands.
- Trackball: the reliable playa choice with gloves, dust, or a fussy touch

screen. Left/right changes focus, up/down uses the current screen's navigation, and click opens or activates the focused item.

- Keyboard: primarily for Claude chat. Enter submits.

On the launcher, left/right moves one tile and up/down jumps one four-tile row. The blue tile is focused. Most app screens put Back at the bottom.

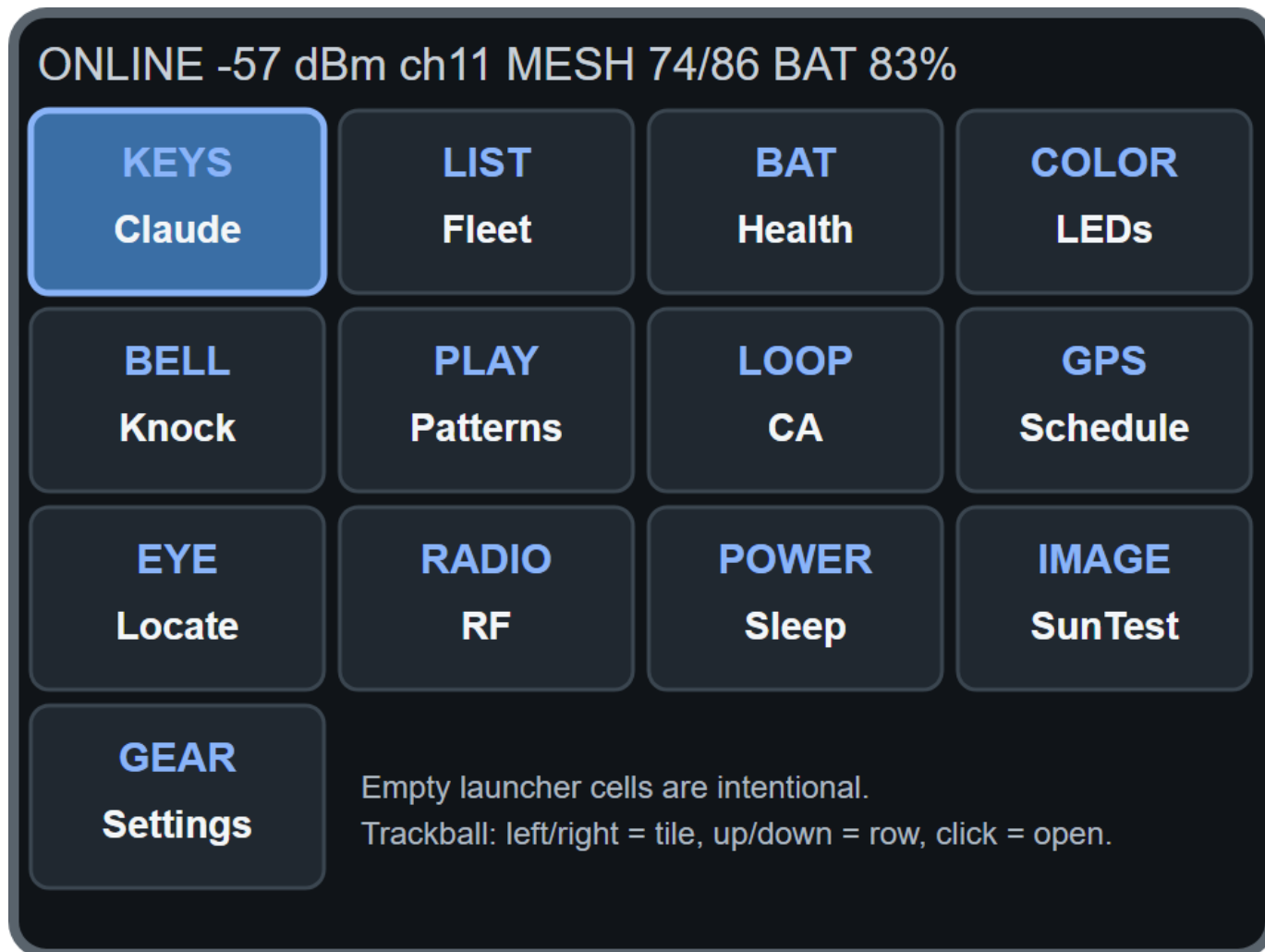


Figure 2. Source-derived launcher map at three times the device's 320 x 240 resolution. It is a faithful screen guide, not a camera photograph.

In direct sun, remove the shipped protective film and tilt the screen so sunlight does not hit it square-on. The panel was judged fairly clear that way. Use the trackball if touch becomes unreliable. T-Deck battery runtime with the census and screen active is still unmeasured, so begin critical work charged and carry USB-C power.

Reading the top status line

The launcher compresses five facts into one line:

```
network  WiFi signal  mesh channel  live/seen fixtures  T-Deck battery
ONLINE   -57 dBm         ch11           74/86                83%
```

Display	Meaning	What to do
ONLINE or WiFi symbol	WiFi is associated on the correct channel; mesh and internet can coexist	Claude should work after time sync
mesh or mesh-only	No infrastructure WiFi, but ESP-NOW is active	Normal for field control without Claude
connecting/refresh	WiFi join is in progress	Wait up to about 15 seconds; mesh returns on timeout
GUARD	The AP channel does not match the stored mesh channel	Fix the AP to channel 11, then retry WiFi
ch11	The stored mesh channel	This is the production value
74/86	74 live now, 86 seen since boot	Only the 74 live fixtures are plausible immediate recipients

Safe control hierarchy

When two instructions disagree, the fixture follows this order:

```
local safety and battery veto
-> transport-dark or active dark lease
-> active direct color or program lease
-> temporary Day Dark / Night Show baseline
-> UTC civil-twilight schedule
-> panel-current fallback
```

This explains several apparently surprising results:

- Night Show cannot defeat low-battery shutdown.
- LED Studio can deliberately light a healthy fixture during scheduled day.
- Dark can suppress a scheduled night show.
- A strike request can be sent successfully while the fixture refuses to move

because its time, power, arming, or mechanism gate says no.

Health: start here

Use Health for the fastest whole-fleet triage. It fits the 141-entry production-health registry on one screen and appends unexpected live IDs.

Tile	Meaning
Green	fresh raw VBAT is above 3.20 V
Yellow	fresh raw VBAT is above 3.10 V and at or below 3.20 V
Red	fresh raw VBAT is at or below 3.10 V

Tile	Meaning
Grey	rostered fixture is not fresh/on-air
Blue	fixture is live, but its battery value is not plausible
Cyan border	live ID is outside the normal production-health roster

These colors are triage bands, not the fixture's complete ADR 0023 power-state machine. Do not force a red fixture to perform merely because it still answers.

Tap or focus-and-click a square to see callsign and short ID, voltage/current, age, RSSI/PDR, supply, advisory SOC, class, program, lifecycle, sensor signature, firmware, registry status, battery capacity, and role.

Grey means not fresh, not automatically broken. After bridge boot, wait at least one normal sleep cadence before calling a field fixture missing. Older or PROTECT firmware can require a 16-minute census before absence becomes strong evidence.

Fleet: inspect and identify individual fixtures

Fleet is a readable list of recent peers. It shows:

- a chip of the fixture's reported rendered color;
- callsign when known, otherwise short ID;
- class letter: D downlight, P perimeter, U uplight, C chandelier;
- age, RSSI, PDR, advisory SOC, and active program.

Open a row for details and Identify, which requests a ten-second green blink. The visible Fleet table is capped at 64 peers and is ordered live-first; it is not a complete 130-fixture roster view. Use Health for full-roster triage and RF for radio summaries.

Callsigns are display and command-entry aliases. Short MAC remains authoritative for flashing, OTA, manifests, persistence, and incident records. Unknown peers fall back to short ID. The special Magic Wand appears as Thor [F40344]; its firmware is protected from ordinary fleet OTA by ADR 0050.

LEDs: simple colors and blinking

Use LEDs to run LED Studio:

1. Select all, downlights, perimeter, uplights, or chandelier.
2. Touch a labeled swatch: red, amber, green, cyan, blue, purple, white, or pink.
3. Set dim.
4. Choose solid or blink 1Hz.
5. Watch the line saying how many fresh fixtures are receiving the stream.
6. Use Stop when finished.

Changing a swatch starts or updates the stream immediately. Class data arrives in full heartbeats and can take about 60 seconds to fill after bridge boot. If a class filter finds unexpectedly few fixtures, wait and check Fleet/Health before assuming those lights are missing.

The stream continues when you press Back. It stops only when you press Stop, start a competing stream such as Patterns, or start a Sleep/Dark action. If the stream disappears, fixtures return to autonomous behavior in about three seconds.

Patterns: deterministic handheld shows

Use Patterns for a manual, microphone-independent look:

- modes: Wash, Chase, Wave, Twinkle;
- palettes: Ember, Forest, Ocean, Aurora, Moon;
- filters: fixture class and stable cohort A-D;
- controls: speed 1-8 and intensity 8-255.

Set the controls, press Start, and confirm the target count. Back leaves the pattern running; Stop ends it. Starting Patterns replaces LED Studio, and starting LED Studio replaces Patterns. They are two faces of one stream owner.

Patterns v1 is in the current T-Deck image and its model is heavily native-tested, but the named HEX/RGBW physical acceptance matrix remains open. Use explicit canaries before an important show. Microphone reactivity is not part of T-Deck Patterns yet.

Sleep / Dark: know the difference

The screen defaults to low-power sleep for 8 hours. Read the selection before applying it.

Action	LEDs	Radio	Can cancel remotely?	Best use
Dark	rail off	awake	Yes, with Release Dark or lease expiry	Short reversible blackout and diagnostics
Low-power sleep	rails off	off	No; timer wake or physical reset/power cycle	Overnight energy saving

Available durations are 10 minutes, 1 hour, 4 hours, 8 hours, and 12 hours. Both actions show live/seen counts and require an on-device confirmation with cancel focused by default.

For Dark, use Release Dark to return reachable fixtures to autonomous behavior. That button cannot wake a sleeping fixture. For Sleep, watch the live count fall after the command; only fixtures that were awake to hear it can leave the census. At timer wake they resume normal behavior automatically.

Dark-but-awake draw has measured roughly 126-144 mA per fixture. Rails-off sleep is sub-mA. Dark is not a power-saving substitute for an overnight sleep.

Schedule: Auto, Day Dark, and Night Show

The Schedule app shows the T-Deck GPS summary and whether a valid UTC fix is available. It offers three field baselines:

- Auto: use UTC civil twilight at Black Rock City, with the fixture's

bounded solar/power fallback if trustworthy time expires.

- Day Dark: temporarily force the daytime baseline.
- Night Show: temporarily force the nighttime baseline.

The choice is RAM-only and clears on fixture reboot. The T-Deck repeats it for six minutes so a fixture on the normal 300-second daytime sleep cadence has a full opportunity to hear it. Direct colors and program leases can override this baseline; dark and local safety remain above it.

Schedule is a production-direction feature with native coverage and first-canary firmware evidence, but the full multi-fixture sleep-cycle acceptance matrix is still open. Verify on named canaries before relying on it fleet-wide.

Knock: request a bamboo strike

The target list contains one row per fresh fixture plus ROLL ALL (targeted). Set a pulse from 5 to 300 ms; 40 ms is the ordinary starting point.

- A single target sends one addressed request.
- Roll All sorts every fresh ID and sends one addressed request every 80 ms.
- Roll All is not synchronized. The display reports sent requests, not

guaranteed mechanical impacts.

Fixtures refuse broadcast strikes. They may also refuse an addressed request at night, on low power, when disarmed, or when the mechanism gate says no. The current full-fleet roll fixes an older 32-fixture limit but still awaits its explicit hardware recheck. Use a named single fixture before using the roll.

CA: cellular-automaton studio

CA Studio leases a program to the fleet. The program selector contains:

UI name	Program	Meaning
idle	0	no active artistic program
ca	1	Greenberg-Hastings cellular automaton
bridge	2	bridge-show program
direct	3	direct-frame program
dark	4	dark program

For CA, the sliders mean:

- K neighbors: excitation threshold;
- spark /256: spontaneous activation probability;
- refractory: refractory ticks;
- tick (ds): tick period in tenths of a second;
- hue: base hue, 0-255.

Choose a 120 s, 300 s, 600 s, or 1 h lease and press Apply. Fleet-wide apply requires confirmation. Release returns to autonomous behavior.

The current fixture needs a release-then-re-lease workaround for parameter changes to the same program. Bridge OS performs it automatically, which can cause one brief visible blip. This is expected until the fixture-side reapply gap is fixed.

RF: read-only radio diagnostics

RF Diagnostics does not send a mesh command. Its links page shows:

- live, seen, stale, roster-unobserved, and foreign-live counts;
- observation coverage;
- strongest and weakest fresh peers by RSSI/PDR/age;
- mesh RX, invalid, and ring-drop counts;
- TX success/failure counts; and

- mesh channel, WiFi state, AP channel, and channel-guard state.

Use the frames button for the newest-first valid-frame tail. PDR - is an honest unavailable value, especially during the first minute; it is not zero. The app is safe for passive diagnosis while another operator owns show control. Its physical two-page acceptance check remains open in the current image.

Claude: ask the bridge in plain language

Claude needs WiFi, valid time, and a provisioned API key. Mesh functions do not. When the WAN is down, a typed message queues in RAM and the app shows that state instead of pretending the mesh failed.

Useful requests include:

```
Which fixtures have been quiet for more than five minutes?
Show me the status of Ponyta.
Identify Ponyta in green for ten seconds.
What are the last ten mesh frames?
Lease CA to the whole fleet for five minutes.
```

The tool surface is intentionally limited to:

- mesh census;
- one-node status;
- identify;
- one-fixture strike;
- temporary program lease; and
- recent frame tail.

Claude cannot represent OTA, reboot, profile changes, lifecycle sleep, battery capacity, or other persistent maintenance operations. A fleet-wide program lease opens the same physical confirmation rail used by local apps. If it times out or is denied, the command is not sent.

Claude's census response is intentionally bounded: it examines up to 64 rows and returns up to 24, marked truncated when appropriate. Use Health or the host dashboard for exhaustive fleet accounting.

Settings, SunTest, and Locate

Settings

Settings changes T-Deck backlight and stored mesh channel. Production is channel

11. A channel change persists, and reboot applies it cleanly. Do not change it in the field without an explicit migration or lab plan.

WiFi credentials, API key, and model are displayed only as provisioning status; secrets are entered over USB serial. The current show command prints an API-key prefix, so do not run it on a projected or public terminal.

SunTest

SunTest forces full backlight and displays contrast bars, text, and color patches. Use it to check readability, stuck display areas, or a questionable backlight. Back returns to the stored backlight level.

Locate

Locate is a placeholder. Active RSSI survey packets exist elsewhere in the system, but Bridge OS does not yet retain and model neighbor reports for this UI. Do not expect a map or survey workflow from this tile.

Part II: T-Deck IT and troubleshooting

Network layout

The T-Deck has one 2.4 GHz radio. WiFi and ESP-NOW can coexist only on one channel. The correct field layout is:

```

Starlink in bypass mode
  |
  v
Beryl AX travel router
  2.4 GHz: fixed channel 11, HT20, WPA2-PSK -> T-Deck + ESP-NOW coexistence
  5 GHz: separate SSID                      -> laptops and phones
  |
  v
Internet / Claude

Fixture fleet -----> ESP-NOW channel 11

```

Do not use the router configuration page as proof of channel. Verify over the air. The complete procedure is [Camp network setup](#).

Serial provisioning

Connect USB serial at 115200 baud. The stored settings live in NVS; no secret is compiled into or committed with Bridge OS.

```

set wifi <ssid> <psk>
set key <anthropic-key>
set model claude-sonnet-5
set channel 11
set bl 200
wifi retry
show

```

Current limitation: the serial parser requires the SSID and password to be one token each. It does not support quotes or spaces. Do not type `set wifi Party In The Woods ...`; it would parse the wrong fields. Use an approved one-token bridge SSID for now, or stop and have the firmware corrected. The fixture maintenance SSID and the T-Deck's simultaneous mesh/internet SSID do not have to be the same network.

Useful read-only service commands:

```

probe      # onboard peripheral report
mem        # heap and PSRAM current/minimum values
peers      # serial census table
t          # one-line JSON bridge telemetry
show       # settings and network state; API-key prefix is visible
emit on    # 1 Hz nb-master/nb-peer lines for host tools
emit off

```

Expected hardware probe evidence on the current T-Deck is 8 MB PSRAM, keyboard, touch, ES7210 at 0x40, and GPS NMEA at 38400 baud.

The serial CLI also has state-changing quick commands. Prefer the physical UI for ordinary operation. If IT work requires serial control, name the exact short ID and follow the same operator discipline:

```

iF2B7DC:10 # identify one fixture
I          # identify all for 8 s
KF2B7DC:40 # one addressed 40 ms strike request
UF2B7DC    # exact-target maintenance request for 35 s; no broadcast form
B600       # fleet dark lease for 600 s
b          # release the fleet dark lease

```

U<ID> only asks the named fixture to enter shared-WiFi maintenance. A laptop must still discover, identity-check, upload, and verify the immutable OTA artifact.

T-Deck symptom guide

Symptom	First check	Likely action
Launcher says GUARD	AP channel versus ch11	Pin the AP's 2.4 GHz radio to channel 11, verify by scan, then wifi retry
WiFi join times out	SSID, password, range, WPA2	Mesh should recover automatically; fix WiFi without changing the mesh channel
Claude queues or stays unavailable	ONLINE, SNTP/time, API key	Run show; restore internet/time/key. Continue with local mesh apps
Mesh live count is zero	channel, RF screen, fixture sleep state	Confirm ch11; wait for a full sleep cadence; move closer; do not assume all fixtures failed
Health is mostly grey after boot	observation time	Wait at least 5 minutes; use 16 minutes for older/PROTECT field firmware
A Health tile is red	raw VBAT at or below 3.10 V	Treat it as a low-power lead; inspect details and charging before asking it to perform
LED Studio says targets, but lights do not change	lifecycle, class data, another publisher	Check Fleet program/output, wait for class heartbeats, stop CoreS3 audio or other frame sources
Look stops when another app starts	stream ownership	LEDs and Patterns replace one another by design; start the desired one again
Look continues after Back	app semantics	Return to LEDs/Patterns and press Stop
Sleeping fixtures ignore Release Dark	radio is off	Wait for timer wake or physically reset/power-cycle an explicitly identified fixture
Knock request produces no sound	local strike gates	Check day/surplus, power tier, solenoid arming, mechanism, and exact target
Touch is unreliable	dust, gloves, glare	Use the trackball; remove protective film and tilt the display
Wrong mesh channel was saved	Settings or serial configuration	Restore 11 and reboot cleanly; coordinate before touching any lab channel
UI behaves strangely after long use	memory/peripheral state	Record mem and probe, then reboot the T-Deck; do not mutate fixtures during diagnosis

When a light is missing

Use this order so a sleeping fixture is not misdiagnosed as dead:

1. Health: is its registry tile grey, or is it completely absent/unexpected?
2. Fleet: was it seen, and how old is the last frame?
3. RF: is the mesh up, and are other fixtures arriving normally?
4. Wait: at least one 300-second field cadence; up to 16 minutes for legacy or PROTECT behavior.
5. Compare peers: if nearby fixtures are also absent, suspect bridge position, antenna obstruction, channel, or a local power event.
6. Physical inspection: panel, cable, enclosure, charge state, and antenna keep-out.

7. USB rescue: only after exact identity and the recovery procedure are ready.

Do not infer that `online=true` from an old dashboard entry proves a fresh rejoin.

Part III: CoreS3 desk bridge

What it is

The CoreS3 is the proven desk/laptop bridge and the current audio-reactive fallback. It stays unassociated from infrastructure WiFi, pins ESP-NOW to channel 11, tracks up to 192 peers, and sends the serial lines understood by the host dashboard and JSONL logger.

One CoreS3 hardware stack can run different, mutually exclusive images:

Image	Purpose	Host behavior
Normal bridge	fleet census, dashboard controls, logging	text nb-* serial protocol
Cambium	Cambium radio modem	binary COBS/CRC protocol; never run the text dashboard against it
Audio, built-in microphones	simple audio-reactive publisher	no external audio module required
Audio, Module Audio	RODE/external analog source	proven preferred CoreS3 audio path

Do not mentally merge the artifacts. Confirm the boot banner and on-screen mode before using the USB port.

Normal dashboard quick start

1. Connect the CoreS3 by its own USB-C port.
2. Confirm the screen says `RESONANCE BRIDGE`, not `RESONANCE AUDIO`.
3. Confirm the boot identity reports channel 11 and serial bridge mode.
4. On the laptop, identify the actual port; never assume yesterday's COM number.
5. Launch:

```
python ops/bench/net_bench_dashboard.py --port COM40
```

Replace `COM40` with the observed port. The known Nevada City CoreS3 is full MAC `44:1B:F6:E3:9F:1C`, short ID `E39F1C`; that identity is more authoritative than its port number.

The dashboard's overview glyph combines:

- center shape: fixture class;
- center fill: battery health;
- top sun/plug: inferred source convention plus charger-input evidence;
- thin top bar: actual reported light color;
- fading tile: late or silent heartbeat;
- persistent checkbox: renewable green location tag.

Source icons are an honest inference, not universal panel-versus-USB detection. Use selected/detail values for exact electrical evidence.

CoreS3 audio quick start

Use the full [CoreS3 audio-reactive how-to](#) for RODE button details and tuning. The short version is:

1. Power the entire stack off.
2. Set Module Audio's physical A/B I2S switch to B.
3. Connect the RODE analog cable to LINE/MIC, not the TRRS headset jack.
4. Start the RODE around gain 10, 75 Hz high-pass, boost off, pad off, safety channel off.
5. Power on and confirm audio MODULE TRS ON.
6. Leave two seconds of ordinary ambient sound for calibration.
7. Confirm intended fixtures appear live, then play the source.
8. Tap the CoreS3 screen to cycle Classic, Ember, Huecycle, and Pulse. Use serial A to pause/resume; serial M also changes mode.
9. Stop or pause this publisher before starting T-Deck LEDs or Patterns.

The bridge sends about 10 direct frames per second. If the bridge pauses, unplugs, or fails, fixtures return to autonomous output after about three seconds. The accepted three-fixture baseline showed live RODE RMS, matched frames on all three perimeter fixtures, and zero send failures or receive drops.

CoreS3 troubleshooting

Symptom	Likely cause	Fix
Dashboard prints garbage or cannot parse	Cambium binary image is installed	Stop the text dashboard and flash/use the normal bridge artifact
CoreS3 says RESONANCE AUDIO when a dashboard bridge was expected	Audio artifact is installed	Use the intended named normal artifact; do not guess from COM number
Display says MODULE TRS, but RMS is exactly 0	Module Audio selector is A	Power the stack fully off, move the selector to B, restart
Display says BUILTIN DUAL MIC	Module not found or built-in artifact installed	Reseat the stack, confirm selector B, and confirm the Module Audio artifact
RMS moves, level is weak	low mic output, wrong jack, distance, pad	Charge/power RODE, use LINE/MIC, raise gain gradually, aim and move closer
Level stays near 100% or RODE peak LED is red	analog overload	Lower gain, move back, then use the -20 dB pad if required
Bridge reacts, fixtures do not	fixtures not live, channel mismatch, lifecycle, old firmware	Check live IDs, channel 11, direct-frame support, and authorized daylight override
Unwanted rumble drives the lights	wind, handling, generator, solenoid	Use directionality and windshield first; then 75 Hz or 150 Hz filtering
CoreS3 screen flickers or stays black	wrong/old image or display initialization issue	Confirm the current PSRAM-framebuffer artifact and boot serial; preserve logs before reflash

For daylight audio tests, an authorized technician may temporarily set each named fixture to forced night in RAM with its own serial N1, then must restore automatic lifecycle with N2. This is a bench procedure, not a casual field shortcut.

Part IV: PUCA performance-audio bridge

Honest status

PUCA is the intended primary performance-audio instrument, but it is not a working Resonance bridge yet. The received factory Eurorack oscillator/effect image is not tree firmware. Do not plug it in expecting the lights to react, and do not copy a CoreS3 binary onto it.

The hardware on hand is:

- PUCA DSP Original Edition with 8 MB PSRAM;
- 6 HP Eurorack expansion;
- 4ms Pod20 Powered case and 45 W brick;
- two knobs, capacitive paw, protected CV and trigger inputs;
- onboard microphones plus WM8978 stereo line input; and
- RODE VideoMic NTG with WS11 windshield.

The intended path is:

```
RODE or mixer -> PUCA audio input -> WM8978 -> local envelope/FFT/onset
               -> ESP-NOW channel 11 -> directed fixture frames
               -> automatic fixture fallback if the stream stops
```

PUCA will be an optional publisher, not a required coordinator.

Safe hardware handling now

Before first Eurorack power:

1. Photograph the ribbon orientation.
2. Confirm the red stripe is at -12 V on the Eurorack bus.
3. Confirm the board is the Original Edition, not Strawberry Edition.
4. Label the exact AUDIO input. Do not confuse it with CV or TRIG.
5. Never feed a speaker output into PUCA.
6. Keep bare-board input below 3.3 V peak-to-peak; attenuate and isolate a hot professional line source.

For the RODE, power it manually; PUCA's line input does not provide the plug-in power that the microphone may use for automatic power sensing. Start flat and conservative, exercise the loudest expected source, and preserve roughly 12-18 dB of digital headroom. Do not run two independent automatic gain loops.

What must pass before field use

- Original-edition identity, USB ID, and MAC recorded;
- ribbon, knobs, paw, CV, and trigger hardware checked;
- exact RODE jack and codec route identified;
- unclipped levels recorded for bowls, violin, singing, and onboard microphones;
- a reproducible firmware/puca_bridge/ artifact created;
- one-fixture direct-frame and three-second fallback proof;

- mixed HEX/RGBW proof;
- packet rate, PDR, CPU load, audio overrun, reset, and multi-hour soak evidence;
- RF/range test in the closed metal Pod20 at intended placement; and
- final RODE/codec gain plus knob/paw functions marked on the field unit.

Until those boxes close, use the CoreS3 audio path.

Future PUCA troubleshooting map

This is a bring-up map, not a claim that firmware exists:

Symptom	Check first
Unit does not power in Pod20	ribbon orientation, -12 V stripe, case power
Factory WiFi AP appears	still running upstream firmware; Resonance runtime must remove the AP
Knobs/paw do nothing	upstream hardware tests and the final documented firmware assignments
RODE is silent	manual mic power, battery, exact AUDIO input, cable, codec LINE/AUX route
Audio clips	RODE output, codec gain, input attenuation, 3.3 Vpp limit
Audio works but mesh does not	Resonance firmware, unassociated radio, fixed channel 11, canonical packet header
Mesh works outside the case but not installed	Pod20 RF shadowing, orientation, placement
Lights remain directed after PUCA stops	stop the test and fix stale-frame/failure behavior before any field use

The detailed bring-up record is [PUCA performance audio bridge](#).

Part V: Common field recipes

Make the whole reachable fleet dark now

Use T-Deck Sleep -> dark (radio awake), select a duration, compare live/seen, and confirm. This is reversible with Release Dark, but it is not low power.

If energy saving matters more than immediate reachability, use low-power sleep instead and accept that it cannot be cancelled over the radio.

Save power overnight

Use T-Deck Sleep -> low-power sleep, choose the intended duration, read the confirmation carefully, and record the pre-command live/seen count. Watch live fixtures disappear. Record any already-silent units separately; the command did not prove they slept.

Find one physical lantern

Open Fleet, select the callsign/ID, and use Identify. If the row is absent, use Health and wait through the sleep cadence before walking to the fixture.

Check whether the network or one fixture is bad

Open RF. If mesh RX and many peers advance, investigate the one fixture. If the whole fleet stops at the same time, check T-Deck channel, guard state, bridge position, and the AP before touching fixtures.

Start a simple handheld look

Open LEDs, select a class or all, choose color and dim, and confirm the target count. Stop CoreS3 audio first. Use Stop when finished.

Start audio-reactive lights

Use the CoreS3 Module Audio recipe. Confirm the T-Deck is not streaming LEDs or Patterns. Confirm the correct CoreS3 artifact, selector B, MODULE TRS ON, live fixture IDs, and a sane level before program audio begins.

Ask Claude which lights need attention

From the T-Deck Claude app:

List fixtures quiet for more than 300 seconds, with callsign, short ID, battery, and last RSSI. Say clearly if the result is truncated.

Then verify the result in Health. Claude is a convenient interface to the same bounded local census, not a separate source of truth.

Prepare an exact-target OTA

1. Declare the operator, source commit, immutable artifact revision/SHA, exact target short MAC, and operation.
2. Ensure the fixture has its production LFP or another proven stable supply for reboot ride-through.
3. Use an exact-target maintenance request, never a fleet broadcast by habit.
4. Let the host tool identity-check and upload the already-built binary.
5. Require a fresh post-job heartbeat, exact revision, survival past the 20-second pending-verify window, and another fresh heartbeat.

An upload acknowledgement is not completion. A cached online flag is not rejoin proof. Never select an artifact by newest timestamp or latest.

Part VI: Shift cards

Field operator handoff

- ☐ T-Deck charged; USB-C power available
- ☐ T-Deck identity 8EB508 confirmed
- ☐ launcher shows channel 11
- ☐ live/seen count recorded
- ☐ Health red/yellow/grey exceptions recorded by callsign and short ID
- ☐ active artistic publisher named: none / T-Deck LEDs / T-Deck Patterns / CoreS3 audio
- ☐ any active Dark, Sleep, Schedule, CA, or program lease recorded with expiry
- ☐ no unplanned fixture remains in forced-night bench state
- ☐ T-Deck returned to trusted storage after shift

IT handoff

- ☐ 2.4 GHz AP verified over the air on channel 11, HT20, WPA2-PSK
- ☐ laptops/phones moved to the separate 5 GHz SSID
- ☐ Starlink/Beryl power and WAN state recorded
- ☐ T-Deck show, probe, and mem captured if diagnosing
- ☐ bridge and fixture identities recorded by MAC, not COM port or IP
- ☐ one operator declared for OTA/USB/NVS work
- ☐ exact artifact revision and SHA recorded before any flash
- ☐ fresh post-pending-verify evidence recorded after OTA
- ☐ logs written to a new path, or intentionally appended to the same logical run

Audio handoff

- ☐ active audio bridge: CoreS3 / PUCA test bench
- ☐ exact firmware mode confirmed from screen/boot banner
- ☐ RODE charged, gain/filter/pad recorded
- ☐ Module Audio selector B confirmed with power off
- ☐ input says MODULE TRS ON; RMS and level are sane
- ☐ one artistic publisher only
- ☐ test fixtures restored to automatic lifecycle
- ☐ stream paused/stopped and three-second fallback observed at end

Glossary

Term	Plain-language meaning
Bridge	A device that observes or temporarily directs the fleet
Callsign	Friendly permanent name used in displays and command entry
Short MAC / short ID	Last three MAC bytes, shown as six hex digits; machine identity
Census	What the bridge has heard since boot, with freshness and radio evidence
Live	Fresh enough to be a plausible immediate command recipient
Seen	Heard at some point since bridge boot; may now be asleep or gone
ESP-NOW	The local 2.4 GHz fleet radio link
Lease	Temporary command that expires without a persistent configuration write
Direct frame	Per-fixture RGBW instruction refreshed several times per second
Stale fallback	Return to autonomous output about three seconds after direct frames stop
Dark	LED rail off while radio remains awake
Sleep	Rails and radio off until timer wake
Guard	T-Deck protection that drops wrong-channel WiFi and preserves the mesh
PDR	Packet delivery ratio; meaningful only with enough observation history
NVS	Persistent settings stored on the device
Canary	An explicitly named test fixture used before a broader action

Maintaining this manual

Update this manual when an app label, safety boundary, field workflow, device status, or troubleshooting conclusion changes. Keep historical build hashes in the log rather than turning this guide into another artifact ledger.

The two SVG figures are intentionally repo-native, diffable screen/role maps. Replace or supplement the launcher map with real device photographs when a clean camera capture is available. A useful photo set is: launcher, Health with mixed bands, Fleet detail with callsign/ID, LED Studio, Sleep confirmation, Schedule GPS status, RF links, RF frames, CoreS3 normal screen, CoreS3 audio screen, and the labeled PUCA front/ribbon/audio connections.

Sources of truth

- [Bridge OS README](#)
- [Bridge OS app roadmap](#)
- [Bridge OS platform decision](#)
- [LED Studio and Sleep / Dark decision](#)
- [UTC and schedule decision](#)
- [CoreS3 bridge README](#)
- [CoreS3 audio-reactive how-to](#)
- [PUCA hardware and bring-up record](#)
- [Camp network setup](#)
- [Firmware artifact handoff](#)
- [Canonical fixture callsigns](#)
- [Project log](#)
- [Open work](#)